

Online Resource 1 — Archival Evidence Inventory and Claim–Evidence Crosswalk

Article title: Auditing reasoning structure beyond claim-level scientific entitlement: structural localization, consequence containment, and boundary-preserving adjudication

Journal: Information Systems Frontiers

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Study: Reasoning Path Audit prospective architectural/adversarial validation

Purpose: Provide a claim–evidence crosswalk and inventory the strongest retained evidence supporting the 38-case analysed set without backfilling, harmonizing, guessing, or inferring unavailable case rows.

1. Claim–evidence crosswalk

This crosswalk maps each load-bearing manuscript claim to the retained archival evidence location used to support it. It is intended as a navigation layer: the manuscript states the scientific interpretation, while the sections below preserve the corresponding evidence granularity, case descriptions, denominator rules, paired contrasts, and documented stability ceiling.

Manuscript claim / result	Main-text location	Supporting OR1 section	Evidence supplied
Analysed evidence set: 38 cases across Zones A–E	Methods §2.11; Results §3.1; Table 2	OR1 §3	Zone counts, purpose, and archival availability for A–E.
Archived C05/C06 worked examples and wording boundary	Methods §2.10; Fig. 3	OR1 §4	Retained C01–C06 case descriptions, including C05 convergence failure and C06 non-structural wording boundary.
Zone-E defect detection/localization: 5/5; false localization: 0/6	Results §3.2	OR1 §6	Named E01–E06 cases plus denominator rule separating five true-defect opportunities from six false-localization

Manuscript claim / result	Main-text location	Supporting OR1 section	Evidence supplied
			opportunities.
Zone-E false overpropagation: 0/6	Results §3.3	OR1 §6	All six mixed-boundary cases offered an opportunity for unjustified consequence propagation.
Zone-D hard-negative safety outcomes: 0/8 across specified error modes	Results §3.4	OR1 §5	Named D01–D08 cases and cumulative hard-negative outcomes.
Dependency/provenance discrimination pairs	Results §3.5	OR1 §7	D06/E05, D07/E04, D04/D08/E02, and D03/E03 contrasts.
Boundary-preserving adjudication and anti-forced-certainty evidence	Results §3.6	OR1 §§4, 6–7	Zone-C wording/closure cases, Zone-E mixed boundaries, and D/E positive-negative pairs.
Architecture stability: 0 new states, 0 new primitives, 0 reopenings	Results §3.7	OR1 §8	Documented stability ceiling across S1–S38 only.
Excluded later historical extension and nonclaims	Limitations; Data and material availability	OR1 §§2, 9–10	Eight later historical cases excluded from the analysed denominator; later AIT-II material not treated as independent validation; explicit nonclaims.

2. Evidence-preservation rule

All retained records are treated as frozen historical evidence. Missing case rows are not backfilled, harmonized, guessed, or inferred from later summaries. Case-specific interpretation is restricted to material actually present in the surviving archive. Eight additional historical cases from a later programme extension are not included in the analysed evidence set because their case-level records are unavailable.

3. Documented validation structure

Zone	n	Purpose	Archival availability
A	12	Endpoint-right / path-wrong	Zone-level count and finding summary retained; individual case labels unavailable

Zone	n	Purpose	Archival availability
B	6	Endpoint-wrong / path-wrong	Zone-level count and finding summary retained; individual case labels unavailable
C	6	Endpoint-wrong / path-locally-plausible	Detailed C01–C06 case summary retained
D	8	Hard negatives	Named D01–D08 cases and safety outcomes retained
E	6	Mixed-boundary discrimination	Named E01–E06 cases, denominator discipline and outcomes retained
Total	38	Analysed prospective challenge set	All analysed cases supported by retained zone- or case-level documentation

Zones A and B: the retained S1–S38 summary preserves zone counts and challenge-family findings but not individual case names or row-level outcomes. This supplement therefore does not invent A/B identifiers. Their contribution is limited to the documented zone-level findings used in the manuscript.

4. Zone C — retained case-level evidence

Case	Challenge	State pattern	Failure/ localization	Outcome
C01	Non-transitive support composition	Local association edges SUPPORTED; global causal/mechanistic composition UNSUPPORTED	INVALID_COMPOSITION; L3/PATH	Preserve local associations; invalidate strong global endpoint
C02	Cumulative weak evidence -> unjustified certainty	Weak-support relations individually SUPPORTED; decisive aggregation/"established" closure UNSUPPORTED	INVALID_COMPOSITION + CLOSURE_OVERREACH; L3/PATH	Preserve weaker supported/plausible claim; reject "established"
C03	Local findings -> population generalization	Subgroup claims SUPPORTED; population-wide claim UNSUPPORTED	INVALID_COMPOSITION + SCOPE_JUMP; aggregation path	Preserve subgroup findings; invalidate population-wide endpoint

Case	Challenge	State pattern	Failure/ localization	Outcome
C04	Valid intermediate result -> premature closure	Compatibility and relative preference SUPPORTED; unique mechanism claim UNSUPPORTED	CLOSURE_OVERREACH; terminal closure segment	Preserve intermediate results; withhold exclusive closure
C05	Supported branches -> invalid global convergence	P1 SUPPORTED; P2 SUPPORTED; uniform global claim UNSUPPORTED	INVALID_COMPOSITION; L4 branch-convergence subgraph	Preserve both conditional branches; invalidate unconditional global claim
C06	Valid evidence chain -> wording-ceiling overreach	Edges/path SUPPORTED to evidential conclusion; strong wording UNSUPPORTED	No RPA structural failure required	Preserve valid path; route wording overreach to claim/closure governance

5. Zone D — hard-negative safety evidence

D01 long but valid reasoning path

D02 unusual but valid inference structure

D03 multiple independent valid paths

D04 multiple admissible alternative explanations

D05 weak but properly qualified conclusion

D06 circular-looking structure with genuine external anchor

D07 shared dataset with explicitly non-independent claims

D08 correctly retained UNDERDETERMINED case

Retained cumulative outcomes: FALSE_LOCALIZATION = 0/8;
 FALSE_STRUCTURAL_DEFECT_INVENTION = 0/8; FALSE_OVERPROPAGATION = 0/8;
 CLAIM_CEILING_VIOLATION = 0/8; FORCED_SINGLETON = 0/8;
 FALSE_ENDPOINT_INVALIDATION = 0/8.

Interpretive boundary: these are hard-negative stress-set outcomes, not population specificity or false-positive-rate estimates.

6. Zone E — mixed-boundary specificity evidence

E01: single genuine local defect inside otherwise valid graph

E02: genuine local defect adjacent to valid multiple-admissible-explanation boundary

E03: one defective branch + one independent valid branch supporting a reduced endpoint claim

E04: hidden dependence in one path while provenance is correctly represented elsewhere

E05: genuine local circular dependency embedded within an externally anchored larger graph

E06: locally NOT_AUDITABLE segment surrounded by auditable evidence and a constrained endpoint

Retained outcomes: GENUINE_DEFECT_CASES = 5; TRUE_DEFECT_DETECTION = 5/5;
TRUE_LOCALIZATION = 5/5; NOT_AUDITABLE_CASES = 1;
CORRECT_NOT_AUDITABLE_ASSIGNMENT = 1/1; FALSE_LOCALIZATION = 0/6;
FALSE_OVERPROPAGATION = 0/6; VALID_REGION_PRESERVATION = 6/6;
CLAIM_CEILING_CORRECTNESS = 6/6.

Denominator rule: E06 is not a true-defect case, so true-defect detection/localization uses 5/5.
False localization and false overpropagation use the six-case set because all six cases can reveal an unjustified localization or consequence beyond the prescribed boundary.

7. Retained positive/negative pairs

Pair	Discrimination	Retained rule
D06 vs E05	anchored cycle vs genuine local epistemic circularity	Cycle shape alone is not circularity; self-entitlement without sufficient independent anchor is failure.
D07 vs E04	disclosed non-independence vs hidden dependence falsely treated as independent	Dependence is not failure; false independence entitlement is.
D04/D08 vs E02	genuine ambiguity vs unsupported forced singleton	Ambiguity is not error; forcing certainty without entitlement is error.
D03 vs E03	multiple valid branches vs one bad + one good branch	One bad branch does not kill the whole graph; consequence is selective.

8. Architecture stability — documented evidence ceiling

Across the analysed S1–S38 record: NEW_CANONICAL_STATE = 0;
NEW_REASONING_PRIMITIVE = 0; ARCHITECTURE_REOPEN = 0. Therefore the strongest documented stability claim is that the frozen reasoning-object and canonical-state kernels survived these 38 cases without expansion. No case-specific claim is made for the excluded historical extension whose case-level records are unavailable.

9. Later AIT-II/RPA material — not independent validation

Later AIT-II/KIF–ECM–RPA records include prospective external/semi-real challenge sequences and a paired shadow trial. These materials share architectural/epistemic ancestry with the present research programme. The paired checkpoint explicitly states that shared ancestry blocks representation of those results as independent external validation. They are therefore not pooled into the AIT1.1 validation denominator and are mentioned only as follow-on development evidence.

10. Explicit nonclaims

population-level accuracy or specificity

independent external validation

inter-adjudicator reliability

universal reasoning-error detection

universal architecture completeness

production reliability

superiority over external reasoning systems

retrospective backfilling of unavailable historical case rows

10. Public reference software

A minimal public reference layer for RPA is hosted at <https://github.com/kensinro/AIDO-RPA>. It provides the manuscript-aligned public method specification, C05 worked example, minimal consequence-containment reference workflow, regression tests, and a public evidence index for the 38-case analysed set. The repository is supplementary to this archival evidence inventory rather than a substitute for it: scientific materiality remains an evidence-constrained adjudicative input, and the repository does not contain private case ledgers, unreleased internal orchestration, or the complete AIDO development codebase.